GENOMELENS

Economic analysis of a genomic report

Cost–utility model of acting on the findings in one whole-genome input, valued at $100,000 per QALY from a healthcare sector perspective.

Source input	**public benchmark genome huDEFDD1**	Input type	**WGS**
Generated	**24 October 2026**	Time horizon	**none: no pooled model ran**
Discounting	**3.0%, costs and QALYs**		

Proprietary.
Demonstration report for a public research genome. GenomeLens, its economic models and its methods are proprietary. This report shows results, not the method: it is not open-source software, not a description sufficient to reproduce the analysis, and not a license to use either.

Public benchmark genome.
A public benchmark genome. It is a real person's genome; the findings are shown to demonstrate the analysis and are not a clinical report.

Research and educational use — not medical or financial advice.
This is a decision-analytic model, not a clinical or economic evaluation submitted to any authority. Its parameters are population averages applied to one genome: it values the expected consequences of acting on a pattern of findings, and does not predict what will happen to any individual. Monetary figures include health valued at a willingness-to-pay threshold and are not cash returned to anyone. Clinically relevant findings require confirmation by an appropriate professional — a physician, clinical pharmacist or board-certified genetic counselor — before anything follows from them.

Economic impact overview

Modeled economic impact

NO HEADLINE VALUE IN THIS GENOME

GenomeLens assessed every finding with an economic pathway. None carries a modeled value; each is classed with its reason in the table of all economically relevant findings, and none is zero by default.

1	3	1
TESTING-STRATEGY / POPULATION-LEVEL VALUE	TRUE ZERO	UNQUANTIFIED

SUPPORTED CALCULATION AT THE ASSUMED AGE

None supported — No decision has every model input sourced at the assumed age, so no value is supported there (an empty set, not a measured zero). This strict section admits only decisions whose every input is sourced at the standardized age of 45; it does not replace the modeled values shown with their own classification.

Research and educational use only; not medical, clinical or financial advice. Standardized economic scenarios: age 45, using genomic findings identified from publicly available research datasets. Genomic calls in this proof of concept have not been independently confirmed by a clinical laboratory and are not diagnostic results.

All economically relevant findings

Every economic result of this genome, ranked within its class. Values in different classes are never added.

FINDING	DECISION PATHWAY	MODELED NMB	EVIDENCE CLASSIFICATION	KEY LIMITATION
Hypothetical, based on assumptions: any adult, before results	Population screening for Lynch syndrome, hereditary breast and ovarian cancer and familial hypercholesterolemia	**per patient tested**	TESTING-STRATEGY / POPULATION-LEVEL VALUE	A population testing-strategy average per patient tested, not this genotype's value
COMT		**valued at zero**	TRUE ZERO	No CPIC guideline doses citalopram or escitalopram by COMT genotype: the CPIC antidepressant guideline's genes are …
Folate Metabolism (MTHFR) — Nutrient-metabolism variant — deficiency prevention		**valued at zero**	TRUE ZERO	The genotype does not change care
ATP7B rs1061472 (a benign polymorphism, not a Wilson disease finding)		**valued at zero**	TRUE ZERO	Not a pathogenic variant. ATP7B c.2495A>G p.Lys832Arg (rs1061472), the only variant behind this row, is classified …
NUDT15 Intermediate Metabolizer (IM) (thiopurines (azathioprine / 6-MP))		**not quantified**	UNQUANTIFIED	The trial-based economic evidence for genotype-guided thiopurine dosing …

State: the figures on this page are model outputs, labeled UNRESOLVED (not established for this donor) unless a row carries another label. The qualified personal medical value is NOT AVAILABLE: no quantity qualifies.

All economically relevant findings (continued)

The evidence hierarchy

- **Evidence-supported**: every model input is sourced and the value does not rest on an age that is stated rather than observed (badge EVIDENCE-QUALIFIED)
- **Primary modeled**: the selected economic model for a decision; it may depend on the stated assumptions shown beside it
- **Once the decision arises**: the value per person if the decision arises; not an expected value today
- **Alternative modeled**: a competing model or sensitivity scenario for the same decision; never added to the primary result
- **Conditional**: valid only if a stated clinical or treatment condition holds; never added
- **Family / cascade-testing**: accrues to relatives or another beneficiary; never in the individual's personal total
- **Population / testing strategy**: a value per person screened or tested in a population, not this donor's value; never added **in this genome**
- **Hypothetical / projected**: an exploratory scenario on stated assumptions, not an observed or established benefit; in no total **in this genome**
- **Negative**: modeled, and the costs exceed the health gain valued at $100,000 per QALY; shown, never hidden
- **True zero**: the evidence supports no incremental benefit or change in care within the defined decision **in this genome**
- **Unquantified**: potentially relevant, but the evidence is insufficient for a monetary value; not $0 **in this genome**
- **Technically unresolved**: the genomic evidence cannot reliably support the interpretation the decision needs; not $0
- **Not applicable**: the model or intervention does not apply to the assessed context (sex, age or setting); not $0

Unknown is not $0. Unquantified is not $0. Technically unresolved is not $0. No supported estimate is not a measured zero. Negative is not zero. An alternative is never added.

Findings valued by their decision models

Net monetary benefit (NMB) is the QALYs gained valued at $100,000 each, less the change in healthcare cost, in 2019 US dollars. It is a health-economic value, not cash saved: a positive NMB can come with higher spending. Each clinical decision is valued once; the categories are never added to each other. Each figure's horizon is its own model's (the carrier models run over a lifetime; no pooled disease model ran for this genome).

Standardized economic scenarios: age 45, using genomic findings identified from publicly available research datasets. Genomic calls in this proof of concept have not been independently confirmed by a clinical laboratory and are not diagnostic results. Age 45 is stated for this run, not observed (one demonstration age for every sample; the PGP public profiles give only an age bracket, and the GIAB and 1000 Genomes samples publish no age). A value whose model depends on age is modeled at that age; a value its source gives for one age only is held at that age, and its row says so. Model evidence is stated apart from whether its context matches the donor, and the qualification stays assumption-driven; an age-dependent value would be evidence-qualified only at an observed or self-reported age.

Sex: known male.

Supported calculation at assumed age 45

None supported AGE 45 ASSUMED No decision has every model input sourced at the assumed age, so no value is supported there (an empty set, not a measured zero). The values limited by inputs besides the age are in B, never added here. This is the stricter test: the modeled values, which rest on stated assumptions, are shown with their own class and are not contradicted by it.

- no decision is supported at the assumed age

A. Evidence-qualified expected value

Not available — no result qualifies; not $0.

- none

B. Other modeled expected value, with further limitations

- none

C. Values that hold only in a stated clinical context

- none for this genome

D. Alternatives and scenarios, shown separately

- none for this genome

E. Actionable pathways not yet quantified

- NUDT15 Intermediate Metabolizer (IM) (thiopurines (azathioprine / 6-MP)) — actionable economically unquantified: the trial-based economic evidence for genotype-guided thiopurine dosing (TOPIC, PMID 26072396; its cost-effectiveness analysis, PMID 30698675) genotyped TPMT only, so its per-carrier value is not transferred to a NUDT15 carrier. The US economic evaluation that includes NUDT15 found in the literature for this report (Fuerte 2025, PMID 39470342) is a 16-week budget impact analysis of preemptive TPMT and NUDT15 testing in children with acute lymphoblastic leukemia, the two genes combined, and a UK cost-utility model (Roberts 2025, PMID 40548539) values testing per patient tested; neither gives an adult carrier's value. CPIC doses NUDT15 intermediate metabolizers with reduced starting doses, as it does TPMT.

F. Valued at zero: care does not change (a result, not a gap)

- **COMT** — valued at zero: no CPIC guideline doses citalopram or escitalopram by COMT genotype: the CPIC antidepressant guideline's genes are CYP2D6, CYP2C19, CYP2B6, SLC6A4 and HTR2A, and COMT is not among them; a multi-gene panel's benefit belongs to the panel's guideline genes (PMID 37032427)
- **Folate Metabolism (MTHFR) — Nutrient-metabolism variant — deficiency prevention** — valued at zero: the genotype does not change care. No change in care (guideline): ACMG 2013, reaffirmed 2020: MTHFR polymorphism testing is not indicated; lowering homocysteine with B vitamins had no vascular benefit (PMID 23288205; PMID 32533132; PMID 20937919).
- **ATP7B rs1061472 (a benign polymorphism, not a Wilson disease finding)** — valued at zero: not a pathogenic variant. ATP7B c.2495A>G p.Lys832Arg (rs1061472), the only variant behind this row, is classified Benign in ClinVar (Variation 35707) and is a common polymorphism, not a Wilson disease allele (ClinVar Variation 35707).

Findings-first economics

No pooled result: no finding enters the pooled model, so there is no combined estimate to state.

This report starts with what was found in the genome. The economics explains the consequence of acting on each finding; the pooled reference case shows what remains after overlapping signals, imperfect adherence and the real cost of acting are all accounted for.

HEALTH GAIN

Not available

no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost

HEALTHCARE COST CHANGE

Not available

no pooled result

MODELED NET BENEFIT

Not available

no pooled result

DECISION UNCERTAINTY

Not available

no pooled result

0

findings carry a standardized economic estimate

8 of the 8 findings shown carry no standardized estimate, and none is assigned one the model cannot support.

NUDT15 Intermediate Metabolizer · COMT COMT Val158Met · Folate Metabolism (MTHFR) · Hereditary Hemochromatosis · Wilson Disease · GABRA2 · CRHR1 — Stress-axis variant · Reproductive genetics

Medication-genotype findings

Where a genotype could change a prescribing decision

These are the clearest cases of a genomic result bearing on a future medication choice. The clinical decision leads; the economics explains why preserving the genotype for that decision has value.

This run produced no medication-genotype finding with a priced economic pathway; a finding whose decision the evidence values at zero is listed with the true zeros.

Decision support, not a directive. Nothing here is an instruction to start, stop or change a medication. Relevance depends on whether the drug is actually prescribed, and the prescribing decision belongs with a qualified clinician who can see the whole clinical picture.

Risk and prevention findings, in detail

No risk or prevention finding of this genome has a standardized estimate, so there is no card to show. The findings are reported without one below, each with its reason on the glance pages; a lower evidence grade changes how a finding should be read, not whether it is shown.

Reported without a standardized value (8): NUDT15 Intermediate Metabolizer (IM)..., COMT COMT Val158Met — COMT Val158Met..., Folate Metabolism (MTHFR)..., Hereditary Hemochromatosis (mild..., ATP7B rs1061472 (benign polymorphism)..., GABRA2 — Alcohol-metabolism..., and 2 more. Each carries its reason on the glance pages (no change in care, not assessable, unquantified, or reproductive information); no value is invented.

How the findings combine

Taken together, what do these findings mean?

There is no pooled reference case for this genome: no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost. Findings bearing on one condition are combined here before any value is credited; with nothing in the pool there is nothing to combine, and no figure stands in for the absent result.

INCREMENTAL HEALTHCARE COST

Not available

no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost

INCREMENTAL QALYS

Not available

no pooled result

REFERENCE-CASE NMB

Not available

no pooled result

CLASSIFICATION

Not available

no pooled result

State: the figures on this page are model outputs, labeled UNRESOLVED (not established for this donor) unless a row carries another label. The qualified personal medical value is NOT AVAILABLE: no quantity qualifies.

How sure is this, and what rests on judgment?

Uncertainty has two layers: the spread of modeled outcomes, and the provenance of the inputs producing that spread. A robust decision does not make its inputs precise, so both are shown.

Net monetary benefit across simulations

No probabilistic analysis of the pooled model: no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost. No probability of cost-effectiveness, interval or one-way swing is stated for it, because there is no pooled strategy to sample. The decision models carry their own parameter uncertainty (2,000 draws from the ranges the sources print, rescaled to a per-carrier value where a source prints another unit); where a decision has a range, its 5th to 95th percentile is printed beside that decision's value.

HYPOTHETICAL: base case and GenomeLens assumption scenario, by finding (hypothetical, based on assumptions)

Base case: the value the published evidence supports for this genome, or “not estimable” with the reason. Scenario: what GenomeLens computes on its own assumptions, graded by the quality of its inputs, with each assumption numbered in the log below. Scenario figures are not from published data for this genome and are not the evidence headline; the reproductive costs belong to a prospective child and are in no personal total. A row valued by its own decision model repeats that model's value as a pointer; the model's basis (published, or hypothetical) is stated beside the headline. This genome's assessment is incomplete (a finding is actionable but unquantified, technically unresolved, or lacks the evidence to value): that is not a conclusion that the genome has no further economic value.

Table 1. Results by finding

FINDING	BASE CASE (PUBLISHED EVIDENCE)	SCENARIO (GENOMELENS ASSUMPTIONS)	WHY THE BASE CASE IS WHAT IT IS
This genome, to the person sequenced (each finding once)	**Not established** Incomplete assessment: not zero	**None projected** Sum of the rows not marked 'not summed': model and scenario values, never this donor's total	**no evidence-supported value; no value from decision models that are not evidence-qualified; no projected finding is summed (a placeholder on its own row is not added); figures marked 'not summed', reproductive values and payer values are excluded**
NUDT15 Intermediate Metabolizer (IM) (thiopurines (azathioprine / 6-MP))	**Not estimable** Not estimable	**No model yet** No model yet	No earlier figure; the decision models' verdict: actionable, with no economic estimate yet: actionable finding with no economic anchor for this gene: the trial-based …
COMT COMT Val158Met — COMT Val158Met — psychiatric medication optimization	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care (a guideline: care does not change) Sources reviewed: PMID 8807664; PMID 33387367; PMID 36509836
Folate Metabolism (MTHFR) — Nutrient-metabolism variant — deficiency prevention	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	No earlier figure; the decision models' verdict: valued at zero: the genotype does not change care Sources reviewed: PMID 17634462; PMID 32533132; PMID 35727271
Hereditary Hemochromatosis (mild variant) carrier (modifier allele)	**Not estimable** Not estimable	**No model yet** No model yet	No earlier figure; the decision models' verdict: no value of its own: it modifies another finding: modifies another finding rather than carrying its own
ATP7B rs1061472 (a benign polymorphism, not a Wilson disease finding)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	No earlier figure; the decision models' verdict: valued at zero: not a pathogenic variant
GABRA2 — Alcohol-metabolism susceptibility variant	**Not estimable** Not estimable	**No model yet** No model yet	An illustrative panel row: shown, not priced
CRHR1 — Stress-axis variant (CRHR1/FKBP5)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care
Reproductive genetics — 2 carrier condition(s) where partner testing would be informative	**Separate arm** Separate arm	**See reproductive table** Reproductive table	Costed per risk in the reproductive rows (a prospective child's value, kept out of the personal total)

Hypothetical, based on assumptions (continued)

HYPOTHETICAL: base case and GenomeLens assumption scenario, by finding (hypothetical, based on assumptions)

Table 2. Reproductive risks identified, with their cost (per pregnancy, for a prospective child; ancestry used: European)

REPRODUCTIVE RISK	CHANCE	COST OF THE CHILD'S CONDITION	EXPECTED COST	TEST THAT RESOLVES IT	SOURCE OF THE COST
HFE H63D Hereditary Hemochromatosis (mild variant); recessive; carrier	**1 in 86** of an affected child, per pregnancy	**Not available** no cost of the child's condition is registered for HFE; only the test that resolves the risk is priced	**—** per pregnancy	**$65** the partner's carrier test	GenomeLens registered estimate
ATP7B K832R Wilson Disease (variable); recessive; common polymorphism, not a disease allele	**not stated** of an affected child, per pregnancy	**Not available** this allele is commonly reported as a polymorphism rather than a disease allele (classified as uncertain significance in the carrier catalog): it cannot complete an affected genotype, so neither a chance of an affected child nor a cost of the child's condition is stated; there is no risk for a partner's test to resolve, so none is priced	**—** per pregnancy	**—** No test priced: no risk to resolve	GenomeLens registered estimate

Methods

How these figures were produced, and how to read them

A decision-analytic model values the expected consequences of acting on each finding in this genome. The figures are model outputs under stated assumptions, kept apart from clinical outcomes.

ANALYTIC FRAME

Perspective	**healthcare sector**
Comparator	**none: no pooled model ran**
Outcomes	**cost, QALYs, NMB**
Time horizon	**none: no pooled model ran**
Discounting	**3%, costs and QALYs**
Willingness to pay	**$100,000/QALY**
Currency	**USD, United States**
Uncertainty	**decision models: 2,000-draw parameter uncertainty; pooled model: no PSA (no pooled model)**

HOW TO READ EACH FIGURE

- **Evidence-supported value:** every model input is sourced and the value does not rest on an age that is stated rather than observed. Where the age is stated, as for every public genome, an age-dependent value is a standardized scenario and is labeled so.
- **Decision-model rows:** computed from published inputs with at least one assumption (for a public genome, the stated age); shown, never presented as evidence-qualified.
- **Hypothetical, based on assumptions:** what a finding could be worth if its curated assumptions held; never added to any counted total.
- **Projected, based on assumptions:** the GenomeLens scenario beside a base case that cannot be computed soundly; not the evidence headline.
- **Reproductive costs:** per pregnancy, for a prospective child, never in a personal total.

Declared limitations

- Population-average parameters are applied to one genome. The model values expected consequences for a person with this pattern of findings, not a prediction for any individual.
- Estimates are illustrative and decision-analytic. This is not a formal economic evaluation or a submission to any authority.
- Where a condition-specific baseline risk is unavailable a generic registered assumption is used; those assumptions dominate the sensitivity analysis and are labeled as such.
- Costs are restated to 2019 US dollars from each source's price year.
- Findings without a modeled decision pathway are excluded from net monetary benefit rather than assigned an invented value.
- Clinically relevant findings require confirmation by an appropriate professional before anything follows from them.

Proprietary method. Demonstration report for a public research genome. GenomeLens, its economic models and its methods are proprietary. This report shows results, not the method: it is not open-source software, not a description sufficient to reproduce the analysis, and not a license to use either.